7-10-25

180

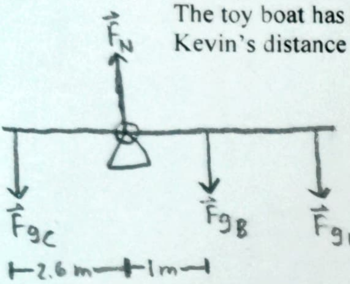
PHY201 Exam on chapter(s): 9, 10

Name:

These questions assess your understanding of the material from Chapters 9, 10 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units.

1. Two children, Charles and Kevin, and a toy boat are balancing on a long seesaw of negligible mass. Charles has a mass of 22 kg and sits 2.6 m away from the fulcrum. Kevin, who has a mass of 17 kg, sits on the other side of the fulcrum, opposite to Charles. The toy boat has a mass of 3.9 kg and is on Kevin's side of the seesaw, a distance of 1.0 m away from the fulcrum. What is Kevin's distance from the fulcrum?

ANSWER: 3.14 m

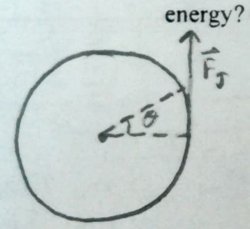


$$\Sigma\tau = \tau_C - \tau_B - \tau_K = 0$$
$$\tau_K = \tau_C - \tau_B$$
$$r_K F_K = r_C F_C - r_B F_B$$
$$r_K m_K g = r_C m_C g - r_B m_B g$$
$$r_K = \frac{r_C m_C - r_B m_B}{m_K}$$
$$r_K = \frac{(2.6\ m)(22\ kg) - (1\ m)(3.9\ kg)}{17\ kg} = \boxed{3.14\ m}$$

$m_C = 22\ kg$
$m_K = 17\ kg$
$m_B = 3.9\ kg$

2. James begins spinning a large grindstone from rest by placing his hand on its edge and exerting a force through part of a revolution. He exerts a force of 151 N through a rotation of 0.38 rad (22°). He exerts the force perpendicular to the grindstone's 48-cm radius and the effects of friction are negligible. If the grindstone's mass is 133 kg, what is its final rotational kinetic energy? Consider the grindstone to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: 27.54 J

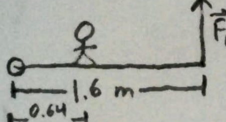


$$\Sigma W = \Delta KE = KE_f - \cancel{KE_i}$$
$$\Sigma W = \Sigma\tau\theta = \tau_J\theta$$
$$KE_f = r_J F_J \theta$$
$$KE_f = (0.48\ m)(151\ \frac{kg\cdot m}{s^2})(0.38\ rad) = \boxed{27.54\ J}$$

$F_J = 151\ N$
$\theta = 0.38\ rad$
$r_J = r_G = 0.48\ m$
$m_G = 133\ kg$

3. Leah is with her daughter, Sophia, at a park. Sophia has climbed onto a merry-go-round. Leah is continuously pushing on the 20-kg merry-go-round with a force of 72 N at its edge, 1.6 m away from the center. Calculate the angular acceleration that is produced in the merry-go-round while 21-kg Sophia is 0.64 m away from its center. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: 3.37 $\frac{rad}{s^2}$



$$\Sigma\tau = \tau_L = I\alpha$$
$$r_L F_L = (\tfrac{1}{2} M_m R_m^2 + M_S R_S^2)\alpha$$
$$\alpha = \frac{r_L F_L}{\frac{1}{2} M_m R_m^2 + M_S R_S^2}$$
$$\alpha = \frac{(1.6\ m)(72\ \frac{kg\cdot m}{s^2})}{\frac{1}{2}(20\ kg)(1.6\ m)^2 + (21\ kg)(0.64\ m)^2} = \boxed{3.37\ \frac{rad}{s^2}}$$

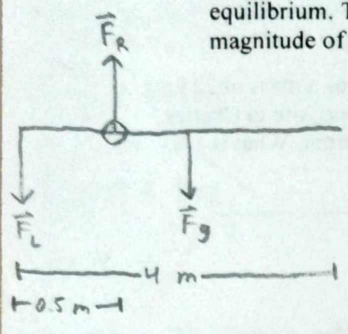
$m_M = 20\ kg$
$F_L = 72\ N$
$m_S = 21\ kg$

+3

$$\Delta x_f = \frac{\cancel{9.8\tfrac{m}{s^2}}\left((0.8\ m)(95\ kg)+(\tfrac{2.3}{2} m)(27\ kg)\right)-(2\ m)(95\ kg+27\ kg)}{\dfrac{(27\ kg)(\tfrac{2.3}{2} m - 2m)(1.2\ m - 2\ m)}{(0.05\ m)(1.2\ m - 2\ m)}} = 0.298\ m = \boxed{29.84\ cm}$$

4. Kevin, a construction worker, holds up a plank of wood 4.0-m long. His left hand pushes an end of the plank down with a force F_L, while his right hand pushes the plank up with a force F_R at a distance of 0.5 m from the left hand. The plank is held in equilibrium. The plank has a mass of 14 kg and its center of gravity is located at the middle of the plank. Calculate the magnitude of force F_L.

ANSWER: 411.6 N



$m_p = 14\ kg$

$\Sigma\tau = \tau_L - \tau_g = 0$

$\tau_L = \tau_g$

$r_L F_L = r_g F_g$

$r_L F_L = r_g m g$

$F_L = \dfrac{r_g m g}{r_L}$

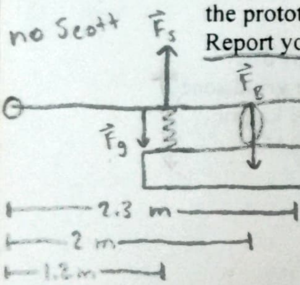
$F_L = \dfrac{(\frac{4}{2} - 0.5\ m)(14\ kg)(9.8\ \frac{m}{s^2})}{0.5\ m} = \boxed{411.6\ N}$

$F_s = k\Delta x$

5. Scott is designing a new 2.3-m-long, 27-kg diving board for DiveMaster, Inc. He bolted a prototype into the ground 2.0 m away from the pool-side end of the diving board with its spring being 1.2 m away from the pool-side end. Scott notes that the spring compresses by 5 cm at equilibrium under the weight of the diving board alone. When Scott, who has a mass of 95 kg, gets on the prototype diving board at a distance of 0.8 m from the pool-side end, by how much does the spring compress at equilibrium? Report your answer in centimeters.

ANSWER: 29.84 cm

no Scott



$m_D = 27\ kg$

$\Delta x_i = 5\ cm$

$\Sigma F_y = F_s - F_g - F_B = 0$

$F_B = F_s - F_g$

$F_B = k\Delta x_i - m_D g$

$\Sigma\tau = -\tau_g + \tau_s - \tau_B = 0$

$\tau_s - \tau_B = \tau_g$

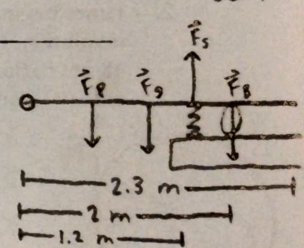
$r_s F_s - r_B F_B = r_g F_g$

$r_s k\Delta x_i - r_B(k\Delta x_i - m_D g) = r_g m_D g$

$r_s k\Delta x_i - r_B k\Delta x_i = r_g m_D g - r_B m_D g$

$k = \dfrac{m_D g(r_g - r_B)}{\Delta x_i(r_s - r_B)}$

Scott



$m_D = 27\ kg$

$m_p = 95\ kg$

$\Sigma F_y = F_s - F_p - F_g - F_B = 0$

$F_B = F_s - F_p - F_g$

$F_B = k\Delta x_f - m_p g - m_D g$

$\Sigma\tau = -\tau_p - \tau_g + \tau_s - \tau_B = 0$

$\tau_s - \tau_B = \tau_p + \tau_g$

$r_s F_s - r_B F_B = r_p F_p + r_g F_g$

$r_s k\Delta x_f - r_B(k\Delta x_f - m_p g - m_D g) = r_p m_p g + r_g m_D g$

$r_s k\Delta x_f - r_B k\Delta x_f = g(r_p m_p + r_g m_D) - r_B m_p g - r_B m_D g$

$\Delta x_f = \dfrac{\cancel{g}(r_p m_p + r_g m_D) - r_B \cancel{g}(m_p + m_D)}{k(r_s - r_B)}$

Great!

ABOVE #4

6. Brandon, a deep-sea fisherman, hooks a big fish that swims away from the boat, pulling the fishing line from the fishing reel. The whole system is initially at rest and the fishing line unwinds from the reel at a radius of 4.8 cm from the reel's axis of rotation. The reel is given an angular acceleration of 77 rad/s² for 1.6 s. How many meters of fishing line come off the reel in this time?

ANSWER: 4.73 m

$\theta_f = \cancel{\theta_i} + \cancel{\omega_i t} + \frac{1}{2}\alpha t^2$

$\Delta x = r\theta_f$

$\Delta x = r\cdot\frac{1}{2}\alpha t^2 = \frac{1}{2} r\alpha t^2$

$\Delta x = \frac{1}{2}(0.048\ m)(77\ \frac{rad}{s^2})(1.6\ s)^2 = \boxed{4.73\ m}$

+3

7. A typical small rescue helicopter has four blades, each of which is 4.2-m long and has a mass of 46 kg. The blades can be approximated as thin rods that rotate about one end of an axis perpendicular to their length, so that the moment of inertia of a blade is $(1/3)ML^2$. The helicopter has a total loaded mass of 1120 kg. Calculate the rotational kinetic energy that the helicopter possesses when its blades rotate at 314 rpm.

ANSWER: 584900.09 J

$KE_{rot} = \frac{1}{2} I \omega^2$

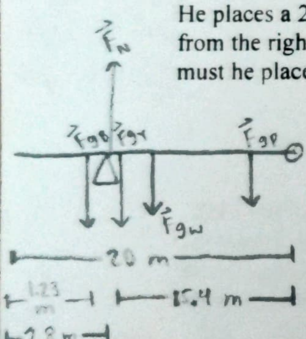
$\frac{314 \text{ rev}}{1 \text{ min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 32.882 \frac{\text{rad}}{\text{s}}$

$KE_{rot} = 4 \cdot \frac{1}{2} \cdot \frac{1}{3} M_b L_b^2 \omega^2$

$KE = \frac{2}{3}(46 \text{ kg})(4.2 \text{ m})^2 (32.882 \frac{\text{rad}}{\text{s}})^2 = \boxed{584900.09 \text{ J}}$

8. Jonathan wants to place three blocks on a 2.6-kg, 20-m-long wooden board to make it balance atop a triangular wedge of wood. He places a 27-kg blue block a distance of 1.23 m from the left end of the board, a 2.2-kg yellow block a distance of 15.4 m from the right end of the board, and places the board on the wedge at a distance of 2.8 m from the left end of the board. Where must he place a 2.7-kg purple block, measured from the right end of the board, in order to balance the board?

ANSWER: 9.9 m



$\Sigma F_y = F_N - F_{gB} - F_{gY} - F_{gw} - F_{gp} = 0$

$F_N = g(m_B + m_Y + m_w + m_p)$

$\Sigma \tau = \tau_p + \tau_w + \tau_Y - \tau_N + \tau_B = 0$

$\tau_p = \tau_N - \tau_w - \tau_Y - \tau_B$

$r_p F_p = r_N F_N - r_w F_w - r_Y F_Y - r_B F_B$

$r_p m_p g = r_N g(m_B + m_Y + m_w + m_p) - r_w m_w g - r_Y m_Y g - r_B m_B g$

$m_B = 27$ kg

$m_Y = 2.2$ kg

$m_p = 2.7$ kg

$m_w = 2.6$ kg

$r_p = \frac{(20 \text{ m} - 2.8 \text{ m})(27 \text{ kg} + 2.2 \text{ kg} + 2.6 \text{ kg} + 2.7 \text{ kg}) - (\frac{20}{2} \text{ m})(2.6 \text{ kg}) - (15.4 \text{ m})(2.2 \text{ kg}) - (20 - 1.3 \text{ m})(27 \text{ kg})}{2.7 \text{ kg}}$

$r_p = \boxed{9.9 \text{ m}}$

9. Aaron flips his bicycle upside down and moves the pedals so that the rear wheel begins spinning. The wheel starts from rest and reaches a final angular velocity of 379 rpm in 6.5 s. Aaron then slams on the brakes to cause an angular acceleration of -21 rad/s². How long does it take the wheel to stop?

ANSWER: 1.89 s

$\cancel{\omega_f} = \omega_i + \alpha t$

$\frac{379 \text{ rev}}{1 \text{ min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 39.689 \frac{\text{rad}}{\text{s}}$

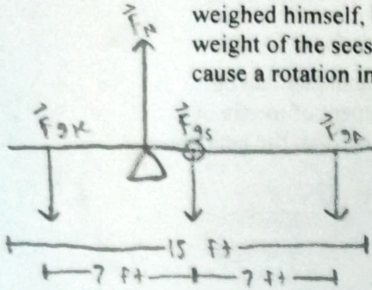
$t = \frac{-\omega_i}{\alpha}$

$t = \frac{-39.689 \frac{\text{rad}}{\text{s}}}{-21 \frac{\text{rad}}{\text{s}^2}} = \boxed{1.89 \text{ s}}$

+3

10. Kevin is building a seesaw for his daughter, Abigail, and plans on making it 15 ft long. He plans to sit on one side, 7.0 ft away from the center of the seesaw, and his daughter will sit on the other side, 7.0 ft away from the center of the seesaw. Kevin has weighed himself, his daughter, and the seesaw and learned that his weight is 218 lbs, his daughter's weight is 37 lbs, and the weight of the seesaw is 12.6 lbs. He wants to place the fulcrum so that the seesaw is perfectly balanced, so that Abigail can cause a rotation in the seesaw just as easily as he can. Where must the fulcrum be placed?

ANSWER: 4.73 ft from center on Kevin's side

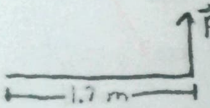


$F_{gK} = 218$ lbs
$F_{gA} = 37$ lbs
$F_{gS} = 12.6$ lbs

$$\Sigma F_y = F_N - F_{gK} - F_{gS} - F_{gA} = 0$$
$$F_N = F_{gK} + F_{gS} + F_{gA}$$
$$\Sigma \tau = -\tau_A + \tau_N + \tau_K = 0$$
$$\tau_N = \tau_K - \tau_A$$
$$r_N F_N = r_K F_K - r_A F_A = r_N(F_K + F_S + F_A)$$
$$r_N = \frac{r_K F_K - r_A F_A}{F_K + F_S + F_A}$$
$$r_N = \frac{(7\text{ ft})(218\text{ lbs}) - (7\text{ ft})(37\text{ lbs})}{218\text{ lbs} + 12.6\text{ lbs} + 37\text{ lbs}} = \boxed{4.73\text{ ft}}$$

11. Mary is with her daughter, Isabella, at a park. She is continuously pushing on a 36-kg merry-go-round with a force of 121 N at its edge, 1.7 m away from the center. Isabella watches excitedly. Calculate the angular acceleration that is produced in the merry-go-round when no one is on it. Consider the merry-go-round to be a uniform disk with negligible friction, so that its moment of inertia is $(1/2)MR^2$.

ANSWER: $3.95 \frac{rad}{s^2}$

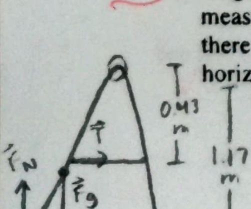


$m_M = 36$ kg
$F_p = 121$ N

$$\Sigma \tau = \tau_p = I\alpha$$
$$r_p F_p = \frac{1}{2} M_m R_m^2 \alpha$$
$$\alpha = \frac{\cancel{r_p} F_p}{\frac{1}{2} M_m R_m^{\cancel{2}}}$$
$$\alpha = \frac{(121 \frac{kg\cdot m}{s^2})}{\frac{1}{2}(36\text{ kg})(1.7\text{ m})} = \boxed{3.95 \frac{rad}{s^2}}$$

12. John places an outdoor sandwich-board advertising sign on the ground. The sign has a taut chain 0.43 m vertically below its hinge. The entire sign's mass is 14 kg. The sign has a vertical height of 1.17 m. The hinge is opened so that the feet of the sign measure 1.5 m apart. Calculate the magnitude of the tension force that the chain exerts on each of the sign's sides, assuming there is no friction between the feet and the sidewalk. (Drawing tip: the sign resembles a large A. The chain resembles the horizontal line in the A and the hinge is at the top of the A.)

ANSWER: 59.83 N



$m = \frac{14}{2}$ kg

$$\Sigma \tau = \tau_T + \tau_g - \tau_N = 0$$
$$\tau_T = \tau_N - \tau_g$$
$$r_T F_T = r_N F_N - r_g F_g$$
$$r_T T = r_N mg - r_g mg = mg(r_N - r_g)$$
$$T = \frac{(\frac{14}{2}\text{ kg})(9.8 \frac{m}{s^2})(\frac{1.5}{2}\text{ m} - \frac{1.5}{4}\text{ m})}{0.43\text{ m}} = \boxed{59.83\text{ N}}$$